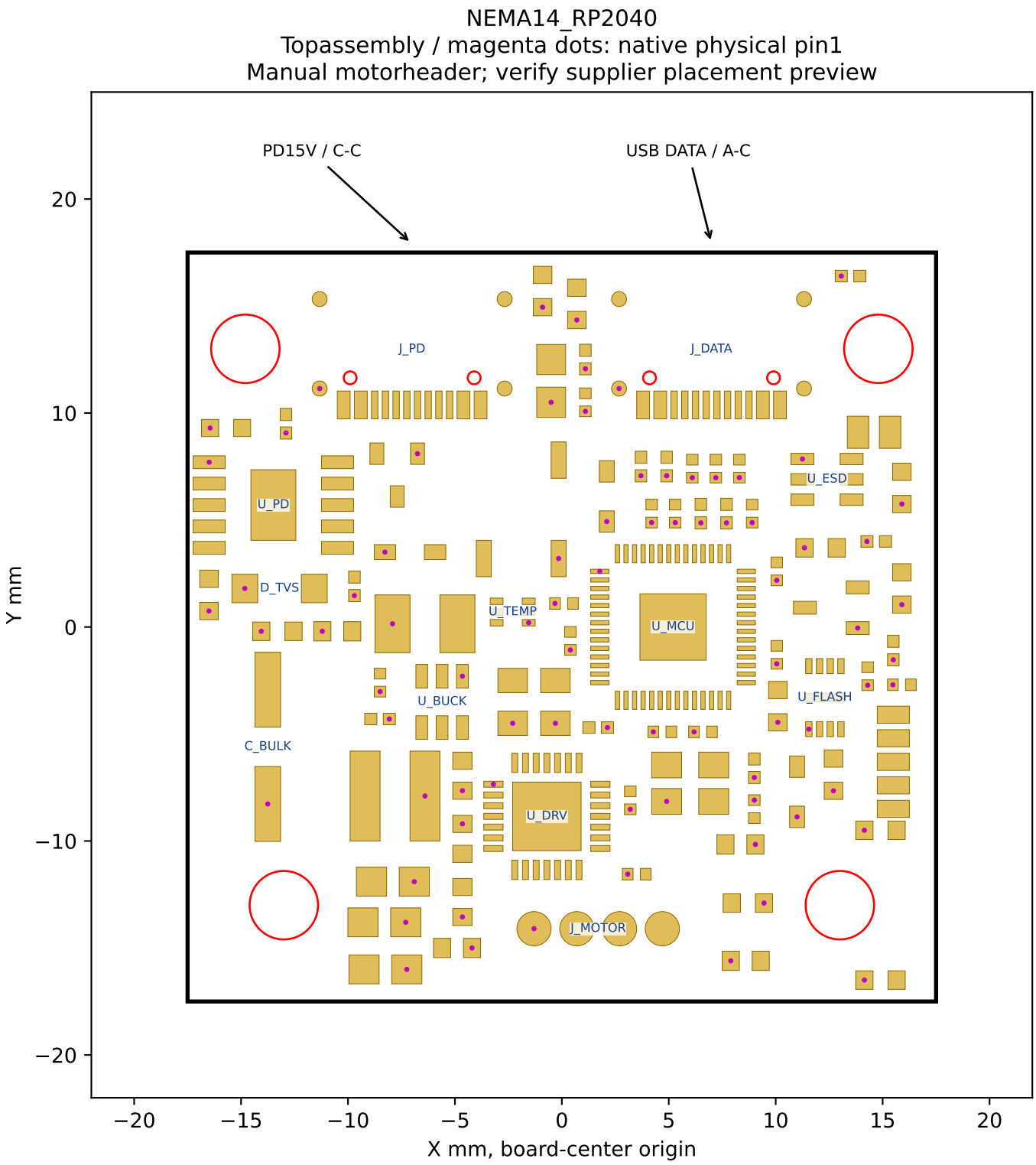



Motor: 1 BlackA+ / 2 GreenA- / 3 RedB+ / 4 BlueB-
PCB NPTH lower26mm / upper29.6mm pitch; front motor26mm pattern is separate.
Cap+ and every diode/IC orientation must match native pads/netlist and supplierpreview.
Filled/capped via list: via_process.csv. See fabrication-notes and mechanical-fit.
SHA256 b28a7426d8bedbb10ae2eef10f6111f28326407e30352b2011bd0beab7ab33bd

# NEMA14_RP2040 - fitted placement reference
## Native source convention; supplier pickup/rotation approval pending

| Ref | Exact MPN | Native center mm | Rotation | JLCPCB |
|---|---|---|---|---|
| J_PD | TYPE-C-31-M-12 | -7.000,12.975 | 180 | C165948 |
| J_DATA | TYPE-C-31-M-12 | 7.000,12.975 | 180 | C165948 |
| U_PD | CH224K | -13.490,5.700 | 270 | C970725 |
| R_PD | ERJPA3F1001V | -13.300,-0.200 | 0 | C441922 |
| C_PD | CL10A105KB8NNNC | -10.500,-0.200 | 0 | C15849 |
| R_PG | 0603WAF1002T5E | -16.500,1.500 | 90 | C25804 |
| U_MCU | RP2040 | 5.200,0.000 | 0 | C2040 |
| C_IO1 | CL05B104KO5NNNC | 1.100,10.500 | 90 | C1525 |
| C_IO10 | CL05B104KO5NNNC | 0.400,-0.650 | 90 | C1525 |
| C_IO22 | CL05B104KO5NNNC | 4.700,-4.900 | 0 | C1525 |
| C_IO33 | CL05B104KO5NNNC | 10.050,-1.300 | 90 | C1525 |
| C_IO42 | CL05B104KO5NNNC | 10.050,2.600 | 90 | C1525 |
| C_IO49 | CL05B104KO5NNNC | 5.300,5.300 | 90 | C1525 |
| C_DV23 | CL05B104KO5NNNC | 6.600,-4.900 | 0 | C1525 |
| C_DV50 | CL05B104KO5NNNC | 4.200,5.300 | 90 | C1525 |
| C_USB | CL05B104KO5NNNC | 6.100,7.400 | 90 | C1525 |
| C_ADC | CL05B104KO5NNNC | 8.900,5.300 | 90 | C1525 |
| C_VREG_IN | CL05A105KA5NQNC | 8.300,7.400 | 90 | C52923 |
| C_VREG_OUT | CL05A105KA5NQNC | 7.200,7.400 | 90 | C52923 |
| U_FLASH | GD25Q16EEIGR | 12.300,-3.300 | 0 | C2986331 |
| C_FLASH | CL05B104KO5NNNC | 14.300,-2.300 | 90 | C1525 |
| R_FLASH_CS | 0402WGF1002TCE | -9.700,1.900 | 90 | C25744 |
| R_BOOT | 0402WGF1001TCE | 14.700,4.000 | 0 | C11702 |
| Y_MCU | ABM8-272-T3 | 6.000,-7.300 | 0 | C20625731 |
| R_XOUT | 0402WGF1001TCE | 9.000,-6.600 | 90 | C11702 |
| C_XIN | 0402CG150J500NT | 3.200,-8.100 | 90 | C1548 |
| C_XOUT | 0402CG150J500NT | 9.000,-8.510 | 270 | C1548 |
| R_RUN | 0402WGF1002TCE | 15.500,-1.100 | 90 | C25744 |
| R_DM | 0402WGF270JTCE | 7.700,5.300 | 90 | C25100 |
| R_DP | 0402WGF270JTCE | 6.500,5.300 | 90 | C25100 |
| R_CC1 | 0603WAF5101T5E | 15.900,6.500 | 90 | C23186 |
| R_CC2 | 0402WGF5101TCE | 13.500,16.400 | 0 | C25905 |
| R_USB_SENSE_H | 0603WAF1002T5E | 12.100,3.700 | 0 | C25804 |
| R_USB_SENSE_L | 0603WAF1002T5E | 15.900,1.800 | 90 | C25804 |

# NEMA14_RP2040 - fitted placement reference
## Native source convention; supplier pickup/rotation approval pending

| Ref | Exact MPN | Native center mm | Rotation | JLCPCB |
|---|---|---|---|---|
| Q_DATA | MMBT3904-7-F | 12.600,0.900 | 0 | C94514 |
| R_VM_H | 0603WAF1003T5E | -0.900,15.700 | 90 | C25803 |
| R_VM_L | 0603WAF1002T5E | 14.900,-9.500 | 0 | C25804 |
| R_VM_BLEED | 0603WAF4700T5E | 12.700,-6.900 | 90 | C23179 |
| C_VM_SENSE | CL05B103KB5NNNC | 15.900,-2.700 | 0 | C15195 |
| U_BUCK | AP63203WU-7 | -5.600,-3.500 | 180 | C780769 |
| L_BUCK | SRN4018-4R7M | -7.800,-7.900 | 180 | C780206 |
| C_BOOTSTRAP | CL05B104KO5NNNC | -8.500,-2.600 | 90 | C1525 |
| C_BUCK_IN | CL32B106KBJNNNE | -6.400,0.150 | 0 | C138687 |
| C_BUCK_OUT1 | CL21A226MAQNNNE | -7.900,-11.900 | 180 | C45783 |
| C_BUCK_OUT2 | CL21A226MAQNNNE | -8.250,-16.000 | 180 | C45783 |
| C_BUCK_OUT3 | CL21A226MAQNNNE | -8.300,-13.800 | 180 | C45783 |
| D_PD | 1N5819WS | 2.100,6.100 | 90 | C191023 |
| D_DATA | 1N5819WS | 11.000,-7.700 | 90 | C191023 |
| U_DRV | A4988SETTR-T | -0.700,-8.850 | 270 | C38437 |
| C_CP | CC0603KRX7R9BB104 | -4.650,-9.900 | 270 | C14663 |
| C_VCP | CC0603KRX7R9BB104 | -4.900,-15.000 | 180 | C14663 |
| C_VREG | CL10B224KA8NNNC | -4.650,-12.850 | 90 | C21120 |
| C_DRV_LOGIC | CL05B104KO5NNNC | 3.500,-11.550 | 0 | C1525 |
| C_VM | CC0603KRX7R9BB104 | -4.650,-6.950 | 90 | C14663 |
| C_VM2 | CC0603KRX7R9BB104 | 8.350,-10.160 | 180 | C14663 |
| C_BULK | EEEFPV101XAP | -13.750,-5.600 | 90 | C178585 |
| R_SA | FRL0805FR240TS | -0.300,-3.500 | 90 | C2930216 |
| R_SB | FRL0805FR240TS | -2.300,-3.500 | 90 | C2930216 |
| R_REF_H | 0603WAF6201T5E | 8.700,-12.900 | 180 | C4260 |
| R_REF_L | ERJPA3F1001V | 14.900,-16.500 | 0 | C441922 |
| C_REF | CL10B103KB8NNNC | 8.600,-15.600 | 0 | C1589 |
| R_ENABLE | 0603WAF1002T5E | 10.100,-3.700 | 90 | C25804 |
| R_SLEEP | 0603WAF1002T5E | 0.700,15.100 | 90 | C25804 |
| J_MOTOR | B4B-PH-K-S(LF)(SN) | 1.700,-14.100 | 0 | C131334 |
| U_ESD | USBLC6-2SC6 | 12.400,6.900 | 270 | C7519 |
| D_TVS | SMF4L16A | -13.200,1.800 | 0 | C1973135 |
| D_STATUS | 19-217/GHC-YR1S2/3T | -15.700,9.300 | 0 | C72043 |
| R_LED | 0402WGF1001TCE | -12.900,9.500 | 90 | C11702 |

NEMA14_RP2040 - fitted placement reference
Native source convention; supplier pickup/rotation approval pending

| Ref | Exact MPN | Native center mm | Rotation | JLCPCB |
|---|---|---|---|---|
| U_TEMP | TMP102AIDRLR | -2.300,0.700 | 0 | C99269 |
| C_TEMP | CL05B104KO5NNNC | 0.100,1.100 | 0 | C1525 |
| R_TEMP_SCL | 0402WGF5101TCE | 3.700,7.500 | 90 | C25905 |
| R_TEMP_SDA | 0402WGF5101TCE | 4.900,7.500 | 90 | C25905 |
| R_TEMP_ALERT | 0402WGF1002TCE | 1.700,-4.700 | 180 | C25744 |
| BZ_ALARM | MLT-5020 | -1.900,5.500 | 90 | C94598 |
| Q_BUZZER | AO3400A | -7.700,7.100 | 90 | C20917 |
| D_BUZZER | 1N5819WS | -7.100,3.500 | 0 | C191023 |
| R_BUZZER_GATE | 0402WGF1000TCE | -8.500,-4.300 | 180 | C25076 |
| R_BUZZER_PD | 0402WGF1003TCE | 1.100,12.500 | 90 | C25741 |
| C_BUZZER | CL21A226MAQNNNE | -0.500,11.500 | 90 | C45783 |